



A Novel Sub-Nyquist FRI Sampling and Reconstruction Method in Linear Canonical Transform Domain

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Abstract

The finite-rate-of-innovation (FRI) sampling frame has drawn a great deal of attention in many applications. In this paper, a novel sub-Nyquist FRI-based sampling and reconstruction method in linear canonical transform (LCT) domain is proposed. First, a new, compact-support sampling kernel is designed to acquire sub-Nyquist samples in time domain, which can be viewed as anti-aliasing prefilter in LCT domain. Then, the corresponding sampling theorem is derived and the reconstruction algorithm is summarized based on annihilating filter and least square method. Moreover, compared with other representative sub-Nyquist sampling methods, the experiment results demonstrate the superior reconstruction performance of the proposed method. The reconstruction ability in noisy environment is also measured by mean square error. Finally, the proposed method is applied to time delay estimation and can obtain super-resolution results.

Keywords Finite of rate innovation · Sub-Nyquist sampling · Linear canonical transform · Time delay estimation

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1 Introduction

In modern signal processing, traditional Shannon–Nyquist sampling theorem is suitable for the perfect reconstruction of bandlimited signal. The sampling rate f_s is equal or greater than the bandwidth B of a signal, that is, $f_s \geq 2B$ [4,24]. However, the bandwidth of signal usually reaches the order of several hundred MHz and even GHz in practical environments, such as radar and communication fields, and such bandwidth demands powerful analog-to-digital converters, burdensome storage costs and consumption. In addition, Shannon–Nyquist sampling theorem fails to work for non-bandlimited signal. Therefore, reducing the sampling rate of signal by exploring novel sampling frame has always been one of the urgent and important research tasks in the field of information engineering.

Recently, sub-Nyquist sampling methods have attracted growing interest in recovering the necessary information effectively, and some popular methods have been applied to many applications [1,2,6,8–11,19,20,25,27,28,30]. The one of classical methods is compressed sensing (CS) [1], which is based on the sparse representation of signal. In the CS frame, however, there are two problems that are commonly encountered. The restricted isometry property is necessary for the observation matrix. Moreover, CS methods suffer from mismatched basis due to discretization grid, which fails to reconstruct the signals or main features perfectly even in the absence of noise [25,27,30]. Another popular method is the FRI-based sampling and reconstruction scheme [6,9,11,16,17,20,21,25,28]. The definition of FRI was first proposed based on the finite degree of freedom of signal over a time interval τ [28]. Generally, for signal model $x(t) = \sum_{l=0}^{L-1} c_l p(t - t_l)$ with FRI $\rho = 2L/\tau$, considering the continuous time Fourier transform, we have $Z(\omega) = \frac{X(\omega)}{P(\omega)}$ and then sample at $\omega = k\omega_0$,

$$Z(k\omega_0) = \frac{X(k\omega_0)}{P(k\omega_0)} = \sum_{l=0}^{L-1} c_l e^{(-j\omega_0 t_l)k} \quad (1)$$

where $P(k\omega_0) \neq 0$ and $k \in \mathcal{K} = [-M, M]$; M is a positive integer. Note that $2M + 1 \geq \rho$ samples are sufficient to reconstruct signal by high-resolution spectral estimation approaches such as annihilating filter (AF) method [22]. The AF annihilates the sequence in (1) as

$$(A * Z)[m] = \sum_{l=0}^L A[l]Z[m - l] = 0 \quad (2)$$

Hence, the time location can be recovered by phase information reconstruction. Other spectral estimation methods such as estimating signal parameter via rotational invariance techniques (ESPRIT) [18] and multiple signal classification (MUSIC) [29] can also be used. The amplitude information can be obtained by least square (LS) method. In the FRI-based sampling method, it is important to design a sampling kernel, which is used to generate sub-Nyquist data and prevent spectral aliasing [26]. In the existing literatures, sampling kernels can be divided into two categories: (1) Sinc-of-class

kernels [16,19,25]: The representative kernels of this type are single sinc function and sum of sinc function (SoS). Particularly, the latter works better in a wide range of engineering fields. (2) Reproducing kernels [6,9,17,20,21]: These include Gaussian kernel, exponential reproducing kernel (E-splines), polynomial reproducing kernel (B-splines) and general reproducing kernel (combination of both E-splines and B-splines). Gaussian kernels [17,28] are not used widely because of the complexity and instability of reconstruction algorithm affected by small values of the Gauss trailer. General reproducing kernel [9] is subject to general strang-fix condition and reconstructs signal via modulated moment, which obtains better reconstruction results than that only E-splines or B-splines. However, the FRI-based methods based on reproducing kernel fail to reconstruct signals with large FRI. Therefore, the SoS sampling kernel has become increasingly popular for researchers, especially those in neuronal signal processing, radar signal processing, super-resolution imaging and ultra-wideband communication [2,8,16,19,25].

However, the above FRI-based sampling theories and methods are concerned only with Fourier transform (FT) domain. In many non-stationary systems and engineering fields, FT is no longer optimal and essential signal processing tool. In [3], a band-limited, single sinc sampling kernel and sampling theorem in fractional Fourier transform (FRFT) domain were presented, but the reconstruction performance and application were not considered. The LCT [13,14,23], as a generalization of the FT and the FRFT [12], can provide more refined features of signal, which has been applied widely to non-stationary signal processing fields. Therefore, it is critical and valuable to develop the FRI-based sampling theory and method associated with the LCT and explore its more potential applications.

In this paper, a novel sub-Nyquist FRI-based sampling and reconstruction method in LCT domain (LCTD) is put forward. First, a newly defined, compact-support and time-limited sampling kernel, the sum of sinc with parameter matrix A (SoS_A), is presented. This SoS_A kernel is able to achieve sub-Nyquist sampling as well as anti-aliasing spectrum in LCTD. Second, the corresponding sampling theorem in LCTD is presented according to sampling procedure. Based on sub-Nyquist samples, the reconstruction algorithm is obtained associated with AF and LS method, and the numerical experiment is implemented to illustrate its technical feasibility. To demonstrate the advantages of proposed method, comparison experiments and mean square error (MSE) are performed, which exhibits better reconstruction performance by the proposed method in LCTD visually and quantitatively. Finally, we apply the proposed method to super-resolution time delay estimation (TDE) for pulse signal and the results demonstrate its potential.

The remainder of this paper is organized as follows. Section 2 is the preliminaries including definition of LCT, linear canonical series (LCS), time-shift property and convolution theorem. Section 3 presents the main method, in which a novel sampling kernel is proposed and FRI-based sampling theorem in LCTD and reconstruction algorithm are derived. Section 4 performs comparison experiments and potential application regarding super-resolution TDE. The conclusion of this paper is obtained in Sect. 5.

Table 1 Some common notations

Notation	Description
$\mathbb{C}$	The set of complex number
$\mathbb{R}$	The set of real number
$\mathbb{N}$	The set of non-negative integers
$\overline{B}$	The complex conjugate of B
$ B $	The modules of B
$*$	The convolution operator of Fourier transform
$*_A$	The convolution operator of LCT with matrix A
$\delta(t)$	The continuous-time Dirac function
$L^2(\mathbb{R})$	The set of square-integrable function on $\mathbb{R}$
$x(t)$	The continuous-time signal with finite energy
$\mathcal{L}_A$	The LCT operator with parameter matrix A
$X_A(u)$	LCT of signal $x(t)$ with matrix A
$\langle f(t), g(t) \rangle$	L^2 -inner integrable between functions $f(t)$ and $g(t)$

2 Preliminaries

This section will first introduce the LCT and LCS including time shift property and convolution theorem in LCTD. Then, the FRI signal model and its representation in LCTD will be presented in the second subsection. Some common notes are shown in Table 1.

2.1 Linear Canonical Transform and Its Series

The LCT [13] of a signal $x(t) \in L^2(\mathbb{R})$ is

$$\begin{aligned} X_A(u) &= \mathcal{L}_A\{x(t)\} = \langle x(t), K_A(t, u) \rangle \\ &= \int_{-\infty}^{+\infty} x(t) \overline{K_A(t, u)} dt \end{aligned} \quad (3)$$

where

$$K_A(t, u) = \begin{cases} \frac{1}{\sqrt{j2\pi b}} e^{-j(\frac{a}{2b}t^2 - \frac{1}{b}ut + \frac{d}{2b}u^2)}, & b \neq 0 \\ \sqrt{d} e^{j\frac{cd}{2}u^2} \delta(t - du), & b = 0 \end{cases} \quad (4)$$

is the transform kernel, $\overline{K_A}$ denotes complex conjugate of K_A , $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is parameter matrix of the LCT satisfying that $|A| = ad - bc = 1$, $a, b, c, d \in \mathbb{R}$.

Particularly, if $A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$, the LCT reduces to the FRFT with rotation angle θ ; if $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, the LCT reduces to the traditional FT. Since the LCT with

$b = 0$ reduces to be a kind of scaling and chirp multiplication operations, this paper will only discuss $b \neq 0$.

If $x(t)$ is periodic signal with period τ , it can be represented by the LCS

$$x(t) = \sum_{m \in \mathbb{Z}} X_A[m] \Phi_A(m, t) \quad (5)$$

where $X_A[m]$ is the LCS of $x(t)$

$$X_A[m] = \langle x(t), \Phi_A(m, t) \rangle = \int_{\tau} x(t) \overline{\Phi_A(m, t)} dt, m \in \mathbb{N} \quad (6)$$

and

$$\Phi_A(m, t) = \sqrt{\frac{j}{\tau}} e^{-j(\frac{a}{2b}t^2 - 2\pi mt/\tau + \frac{d}{2b}(2\pi bmt/\tau)^2)} \quad (7)$$

If matrix A is chosen on purpose, signal $x(t)$ can also be represented by corresponding Fourier series (FS) and fractional Fourier series (FRFS).

Moreover, let us recall two useful properties of the LCT in this paper.

Lemma 1 Time shift property [14] Let $x(t) \in L^2(\mathbb{R})$ be a complex signal, and parameter matrix is $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. if $X_A(u) = \mathcal{L}_A\{x(t)\}$, then

$$\mathcal{L}_A\{x(t - t_0)\} = e^{jcut_0 - jact_0^2/2} X_A(u - at_0) \quad (8)$$

Lemma 2 Convolution Theorem [5] Let $x(t), g(t) \in L^2(\mathbb{R})$ be complex signals, and parameter matrix is $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $X_A(u) = \mathcal{L}_A\{x(t)\}$, $G_A(u) = \mathcal{L}_A\{g(t)\}$, then

$$\mathcal{L}_A\{x(t) *_A g(t)\} = \overline{\lambda_{d,b}(u)} X_A(u) G_A(u) \quad (9)$$

where $\lambda_{d,b}(u) = e^{j\frac{d}{2b}u^2}$ and $*_A$ is convolution operator in LCTD defined as

$$x(t) *_A g(t) = \frac{1}{\sqrt{j2\pi b}} \overline{\lambda_{a,b}(t)} \cdot \{[x(t)\lambda_{a,b}(t)] * [g(t)\lambda_{a,b}(t)]\} \quad (10)$$

where $\lambda_{a,b}(t) = e^{j\frac{a}{2b}t^2}$, and $*$ is convolution operator in FT domain (FTD).

2.2 FRI Signal and Its Representation in LCTD

We are interested in a periodic stream of Dirac impulse

$$x(t) = \sum_{n \in \mathbb{Z}} \sum_{l=0}^{L-1} c_l \delta(t - t_l - n\tau) \quad (11)$$

where τ is the period and L is the number of impulse, $\{c_l\}_{l=0}^{L-1}$ are complex amplitudes and $\{t_l\}_{l=0}^{L-1}$ are time locations. $\{c_l, t_l\}_{l=0}^{L-1}$ are sparsely located in the $\mathbb{C} \times [0, \tau)$.

Let $D(t) = \sum_{n \in \mathbb{Z}} \delta(t - n\tau)$ be a τ -periodic pulse comb, (11) can be rewritten as

$$x(t) = \sum_{l=0}^{L-1} c_l D(t - t_l) \quad (12)$$

From (12), it can be found that $x(t)$ is sparsely represented by $\{c_l, t_l\}_{l=0}^{L-1}$, which is a FRI signal with $\rho = 2L/\tau$ [28].

Therefore, based on Poisson summation formula, $D(t)$ can be expressed by the LCS

$$D(t) = \frac{\sqrt{j2\pi b}}{\tau} \sum_{m \in \mathbb{Z}} \delta_A[mu_0] e^{-j[\frac{a}{2b}t^2 + \frac{d}{2b}(mu_0)^2] + jmu_0t/b} \quad (13)$$

where $u_0 = 2\pi b/\tau$.

Let $D_l(t)$ be $D(t)$ with t_l , using (8) then

$$\begin{aligned} D_l(t) &= \sum_{m \in \mathbb{Z}} \mathcal{L}_A\{\delta(t - t_l)\}|_{u=mu_0} \Phi_A(m, t) \\ &= \frac{1}{\tau} \sum_{m \in \mathbb{Z}} e^{j\frac{a}{2b}(t_l^2 - t^2) + jmu_0(t - t_l)/b} \end{aligned} \quad (14)$$

Further, (12) can be rewritten as

$$\begin{aligned} x(t) &= \sum_{l=0}^{L-1} c_l D_l(t) = \sum_{l=0}^{L-1} c_l \sum_{n \in \mathbb{Z}} \delta(t - t_l - n\tau) \\ &= \sum_{l=0}^{L-1} c_l \left(\frac{1}{\tau} \sum_{m \in \mathbb{Z}} e^{j\frac{a}{2b}(t_l^2 - t^2) + jmu_0(t - t_l)/b} \right) \\ &= e^{-j\frac{a}{2b}t^2} \sum_{m \in \mathbb{Z}} \frac{1}{\tau} \underbrace{\left(\sum_{l=0}^{L-1} \underbrace{c_l e^{j\frac{a}{2b}t_l^2}}_{q_l} \underbrace{e^{-jmu_0t_l/b}}_{u_l^m} \right)}_{Q[m]} e^{jmu_0t/b} \end{aligned} \quad (15)$$

From (15), we have

$$Q[m] = \sum_{l=0}^{L-1} q_l u_l^m \quad (16)$$

with finite rate of innovation $\rho = 2L/\tau$ clearly, where $q_l = c_l e^{j\lambda_a b t_l^2}$ and $u_l = e^{-j\mu_0 t_l/b}$. (16) shares the form of (1) by using LCS. Next, there is a need to design a novel sampling kernel in LCTD to obtain sub-Nyquist samples and recover signal by appropriate reconstruction algorithm.

3 Proposed Method

This section will propose an FRI-based sub-Nyquist sampling and reconstruction algorithm in LCTD. A novel, compactly support sampling kernel will be first designed, which can also be viewed as an anti-aliasing, low-pass filter in LCTD. Then, sub-Nyquist sampling theorem and reconstruction algorithm will be summarized. Finally, numerical experiment will be carried out to verify the technical feasibility of the proposed method.

3.1 FRI-Based Sampling in LCTD

To obtain sub-Nyquist samples of a signal, a sampling kernel, sum of sinc class with a parameter matrix A (SoS $_A$) in LCTD, is designed

$$H(u) = \lambda_{d,b}(u) \sum_{m=-M_A}^{M_A} \text{sinc}\left(\frac{u}{u_0} - m\right) \quad (17)$$

where u is the frequency in LCTD, parameter matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $|A| = 1$, $a, b, c, d \in \mathbb{R}$, $\lambda_{d,b}(u) = e^{j\frac{d}{2b}u^2}$ and $\text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$, $u_0 = 2\pi b/\tau$, $[-M_A, M_A]$ is certain set of sampling sequence.

The corresponding expression in time domain is

$$h(t) = \text{rect}\left(\frac{t}{\tau}\right) \overline{\lambda_{a,b}(t)} \sum_{m=-M_A}^{M_A} e^{jmu_0t/b} \quad (18)$$

where

$$\text{rect}\left(\frac{t}{\tau}\right) = \begin{cases} 1, & -\frac{\tau}{2} \leq t \leq \frac{\tau}{2} \\ 0, & \text{otherwise.} \end{cases} \quad (19)$$

and $\lambda_{a,b}(t) = e^{j\frac{a}{2b}t^2}$.

To visually describe the new sampling kernel SoS $_A$, it is plotted in Fig. 1 with $M_A = 5$. The sampling kernel in (18) is depicted in Fig. 1a, which is compact support over $[-\tau/2, \tau/2)$. Figure 1b shows the corresponding form in LCTD, which can be viewed as low-pass filter. This sampling kernel is able to nullify the aliasing at frequencies $\{mu_0\}$, $m = -M_A, \dots, M_A$ when the sampling rate is chosen appropriately. Figure 2 gives the general sampling procedure: For the inputted continuous-time signal of $x(t)$, it is first filtered by $h(t)$ to obtain a prefiltered version $y(t) = x(t) * h(-t)$. Then, $y(nT)$ is obtained by sampling setup in uniform period T . This sampling procedure and result can be summarized as in detail Theorem 1.

Theorem 1 Let $x(t)$ be a τ -periodic stream of Dirac impulses in (11) with FRI $\rho = 2L/\tau$. Let the sampling kernel SoS $_A$, $h(t)$, be an ideal low-pass filter with frequency width $[-B\pi, B\pi]$ in LCTD, where $B \geq \rho$. If the filtered version of $x(t)$, i.e., $y(t) =$

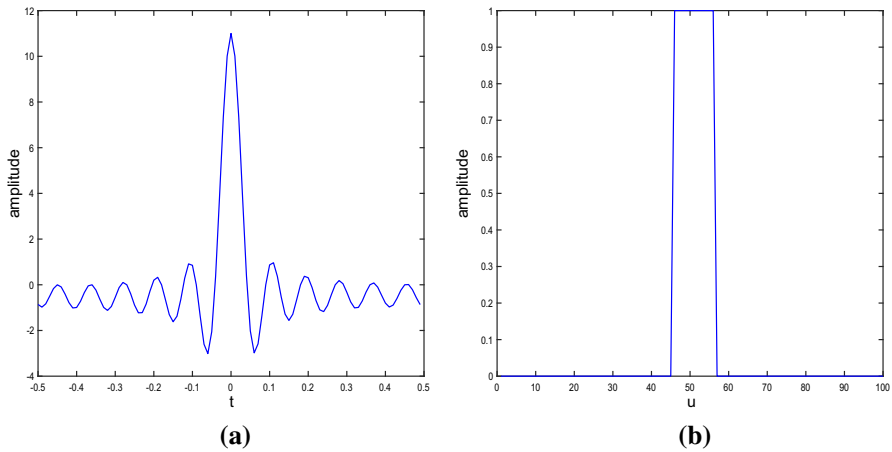
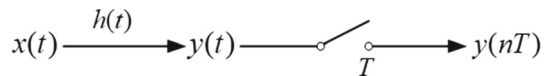


Fig. 1 Sampling kernel SoSA **a** in time domain, **b** in LCTD

Fig. 2 Sampling procedure



$x(t) * h(-t)$ is sampled uniformly at locations $t = nT, n = 0, \dots, N - 1$, then

$$y(nT) = x(t) * h(-t)|_{t=nT}, n = 0, \dots, N - 1 \quad (20)$$

which contains sufficient characterization of $x(t)$ if $N \geq 2M_A + 1$ and $M_A = \lfloor B\tau/(2b) \rfloor$.

Proof The sampling kernel in (18) is compactly supported over $[-\tau/2, \tau/2]$, and the corresponding expression in (17) is low-pass filter in LCTD. Thus, prefilter and sampling can be performed simultaneously. The sub-Nyquist sampling result is shown in (21) by uniform sampling period T . $\square$

3.2 Reconstruction Algorithm

After performing sub-Nyquist sampling of signal $x(t)$, a reconstruction algorithm should also be considered. It is known that $Q[m] = \sum_{l=0}^{L-1} q_l u_l^m, m \in \mathbb{Z}$ has the form of (1) with $q_l = c_l e^{j\lambda_{a,b} l}$ and $u_l = e^{-ju_0 l/b}$. For the FRI-based sampling and reconstruction frame, there are some spectral estimation methods such as Prony's AF method [22], ESPRIT [18], and MUSIC [29]. We utilize AF to reconstruct time location and LS method to retrieve amplitude. The reconstruction algorithm is summarized in Algorithm 1, which certifies the feasibility of Theorem 1. For annihilating step in Algorithm 1, (22) can be rewritten with $A[0] = 1$

$$\begin{aligned}
y(nT) &= x(t) *_{\mathcal{A}} h(-t)|_{t=nT} = \frac{1}{\sqrt{j2\pi b}} \overline{\lambda_{a,b}}(nT) [(x(t)\lambda_{a,b}(t)) * (h(-t)\lambda_{a,b}(t))]|_{t=nT} \\
&\stackrel{a_1}{=} \frac{1}{\sqrt{j2\pi b}} \overline{\lambda_{a,b}}(nT) \langle x_A(t), h_A(t - nT) \rangle \\
&\stackrel{a_2}{=} \frac{\overline{\lambda_{a,b}}(nT)}{\sqrt{j2\pi b\tau}} \left\langle \sum_{m \in \mathbb{Z}} Q[m] e^{jmu_0 t/b}, \text{rect}\left(\frac{t - nT}{\tau}\right) \sum_{m=-M_A}^{M_A} e^{jmu_0(t-nT)/b} \right\rangle \\
&\stackrel{a_3}{=} \frac{\overline{\lambda_{a,b}}(nT)}{\sqrt{j2\pi b\tau}} \sum_{m \in \mathbb{Z}} Q[m] \int_{-\infty}^{+\infty} e^{jmu_0 t/b} \text{rect}\left(\frac{t - nT}{\tau}\right) \sum_{m=-M_A}^{M_A} e^{-jmu_0(t-nT)/b} dt \\
&= \frac{\overline{\lambda_{a,b}}(nT)}{\sqrt{j2\pi b\tau}} \sum_{m \in \mathbb{Z}} Q[m] \int_{-\tau/2}^{+\tau/2} e^{jmu_0(t+nT)/b} \text{rect}\left(\frac{t}{\tau}\right) \sum_{m=-M_A}^{M_A} e^{-jmu_0 t/b} dt \\
&= \frac{\overline{\lambda_{a,b}}(nT)}{\sqrt{j2\pi b\tau}} \sum_{m \in \mathbb{Z}} Q[m] e^{jmu_0 nT/b} \times \sum_{m=-M_A}^{M_A} \text{sinc}\left(\frac{u}{u_0} - m\right) \\
&= \frac{\overline{\lambda_{a,b}}(nT)}{\sqrt{j2\pi b\tau}} \sum_{m=-M_A}^{M_A} Q[m] e^{jmu_0 nT/b}
\end{aligned} \tag{21}$$

a_1 : $x_A(t) = x(t)\lambda_{a,b}(t)$, $h_A(t) = h(t)\lambda_{a,b}(t)$

a_2 : by (15), (18)

a_3 : L^2 – inner product

Algorithm 1 Reconstruction for FRI signal in LCTD

Input:

The continuous time signal: $x(t)$

Ensure:

1. Design sampling kernel $h(t)$ to obtain samples: $y(nT) = x(t) *_{\mathcal{A}} h(-t)|_{t=nT}$ as shown in (20).
2. Calculate $2M_A + 1$ LCS coefficients $Y(mu_0)$ and then obtain $Q[m]$, $m = -M_A, \dots, M_A$.
3. Annihilate $Q[m]$ by AF $A[l]$

$$(A * Q)[m] = \sum_{l=0}^L A[l] Q[m - l] = 0 \tag{22}$$

4. Reconstruct time location $\hat{t}_l$ by the roots of (22) that is $\hat{u}_l = e^{-jmu_0 \hat{t}_l/b}$.

5. Estimate amplitude $\hat{c}_l$ by LS method.

Output:

Reconstructed signal $x(\hat{t})$ with time location $\hat{t}_l$ and amplitude $\hat{c}_l$.

$$\begin{bmatrix} Q[L-1] & Q[L-2] & \cdots & Q[0] \\ Q[L] & Q[L-1] & \cdots & Q[1] \\ \vdots & \vdots & \ddots & \vdots \\ Q[N-1] & Q[N-2] & \cdots & Q[N-L] \end{bmatrix} \begin{bmatrix} A[1] \\ A[2] \\ \vdots \\ A[L] \end{bmatrix} = - \begin{bmatrix} Q[L] \\ Q[L+1] \\ \vdots \\ Q[N] \end{bmatrix} \tag{23}$$

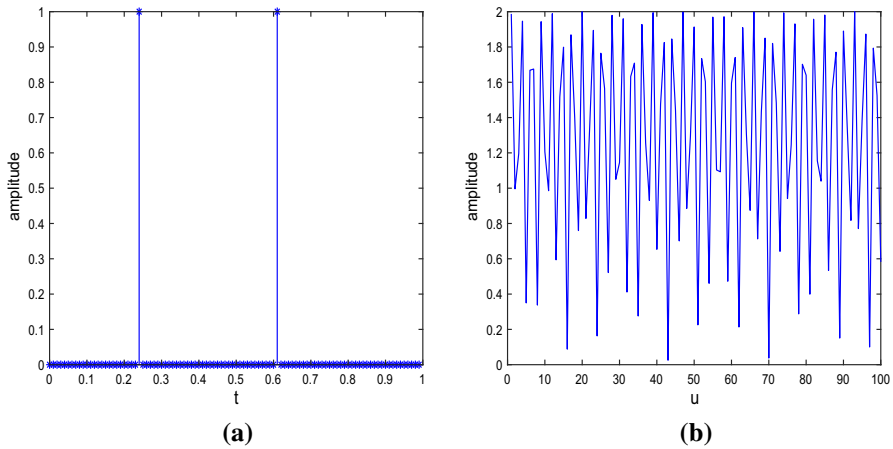


Fig. 3 $L = 2$ Diracs stream with $\tau = 1$. **a** in time, **b** in LCTD

In this system, a solution $A[l]_{l=1}^L$ can be determined uniquely for the given signal, from which the time location t_l can be recovered. As for the amplitude c_l of signal, we have

$$\begin{bmatrix} 1 & 1 & \cdots & 1 \\ u_0 & u_1 & \cdots & u_{L-1} \\ \vdots & \vdots & \ddots & \vdots \\ u_0^{L-1} & u_1^{L-1} & \cdots & u_{L-1}^{L-1} \end{bmatrix} \begin{bmatrix} e^{j\frac{a}{2b}t_0^2} & & & \\ & e^{j\frac{a}{2b}t_1^2} & & \\ & & \ddots & \\ & & & e^{j\frac{a}{2b}t_{L-1}^2} \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{L-1} \end{bmatrix} = \begin{bmatrix} Q[0] \\ Q[1] \\ \vdots \\ Q[L-1] \end{bmatrix} \quad (24)$$

(24) can be solved by the LS method to reconstruct amplitude of signal accurately in the mean-square error (MSE) sense. Therefore, we finish the reconstruction of signal at sub-Nyquist sampling rate in LCTD.

3.3 Numerical Simulation

A signal is produced by a stream of two Dirac impulses with a period $\tau = 1$ and shown in Fig. 3a. The time locations are randomly in $(0, 1)$ and amplitudes $c_1 = c_2 = 1$. Figure 3b shows the spectrum of $x(t)$ in LCTD.

In Fig. 4a, the smooth blue curve is the prefiltered version $x(t) * h(-t)$ of signal and the sub-Nyquist samples (red circle) are obtained via proposed sampling kernel SoS_A. According to Theorem 1, $2M_A + 1 = 5$ samples are sufficient to reconstruct the original signal. The low-pass approximation in LCTD is illustrated in Fig. 4b. According to Algorithm 1, Fig. 5 presents good reconstruction result, from which it is easily found that our proposed method is reasonable and effective for the reconstruction of non-bandlimited signal.

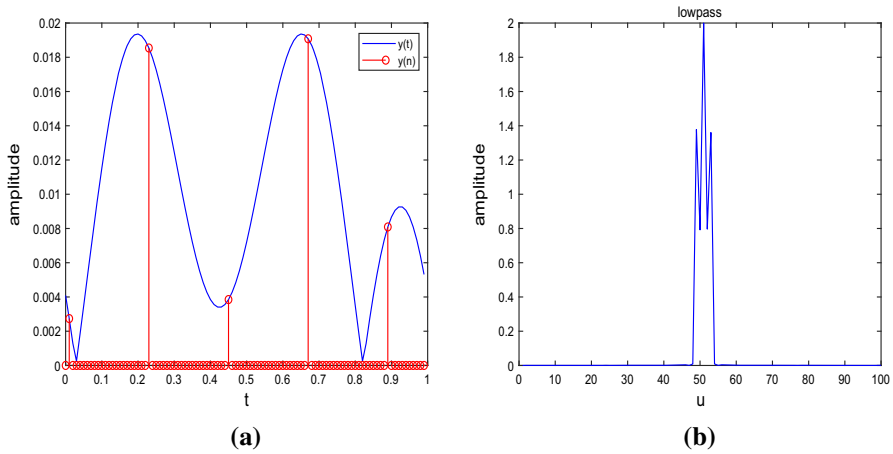
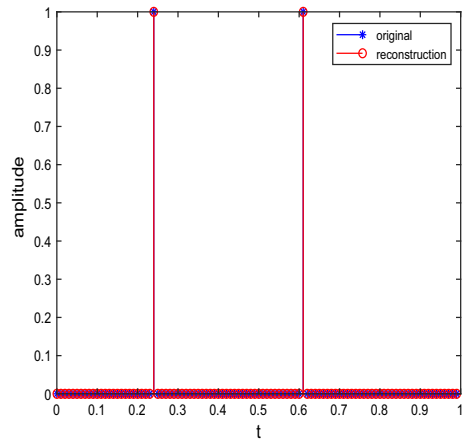


Fig. 4 Sampling results **a** sampling in time, **b** low-pass data in LCTD

Fig. 5 Reconstruction result



4 Comparison and Potential Application

This section will first make some contrastive experiments to verify reconstruction performances of different representative sub-Nyquist sampling methods including CS [30], three FRI-based sub-Nyquist sampling methods in FTD [9,16,20]. Then, the induced reconstruction errors will be measured by MSE under different Signal to Noise Rate (SNR). Finally, the potential application of proposed method on super-resolution TDE will be carried out.

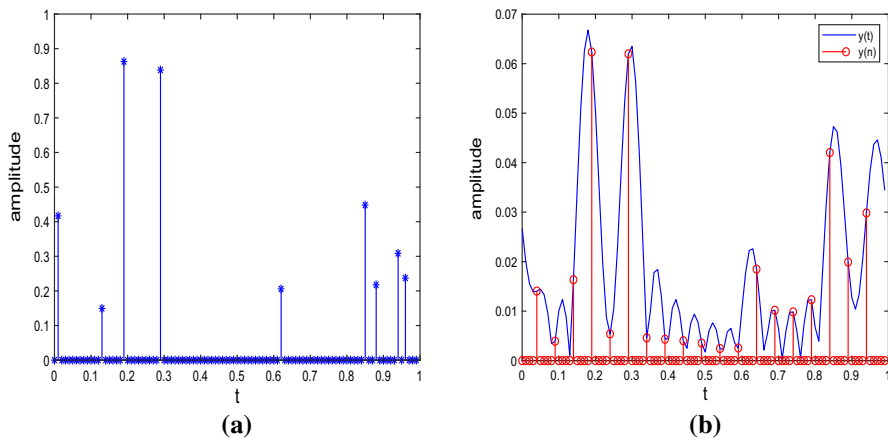


Fig. 6 Signal and sub-Nyquist samples **a** signal, **b** samples with $2M_A + 1$

4.1 Comparison Experiments

Experiment 1: A stream of $L = 9$ Dirac impulses is generated, and both the time locations and amplitudes are randomly in $(0, 1)$,

$$x(t) = \sum_{l=0}^{L-1} a_l \delta(t - t_l) \quad (25)$$

which is plotted in Fig. 6a. Its prefiltered version and sub-Nyquist sampling result are depicted in Fig. 6b, in which the smooth blue curve is prefiltered FRI signal and red circles are the $2M_A + 1$ samples obtained by proposed FRI-based sub-Nyquist sampling method. Three other sub-Nyquist sampling methods are also executed to reconstruct above signal.

Figure 7a, b shows the results obtained by CS with $M/4$ and $M/3$ data, where M is the signal length. From Fig. 7a, many time locations cannot be recovered well and large amplitude errors are generated due to poor signal information ($M/4$). Figure 7b has better reconstruction performance than Fig. 7a due to the use of more data. If one wants better reconstruction results, more data and even full data of signal are needed, which will suffer from much power consumption and high hardware requirement. For the FRI-based sub-Nyquist sampling methods, different sampling kernels are considered, among which Fig. 8a, b is the results based on the E-splines and general reproducing kernel, respectively. From Fig. 8, it can be validated that the method based on reproducing kernel fails to reconstruct signal for large number of impulse, which means that the reconstruction method based on this kind of reproducing kernel is difficult to utilize into practice. Figure 9a, b shows reconstruction results based on SoS-class sampling kernel in FTD and LCTD, respectively, in which $b = 1$ in parameter matrix A . It is easily found that the proposed approach is able to reconstruct signal perfectly. Comparing Fig. 7 with Fig. 8, FRI method based on SoS-class sampling kernel has superior reconstruction ability and stability. Although the results in Fig. 9 seem to have

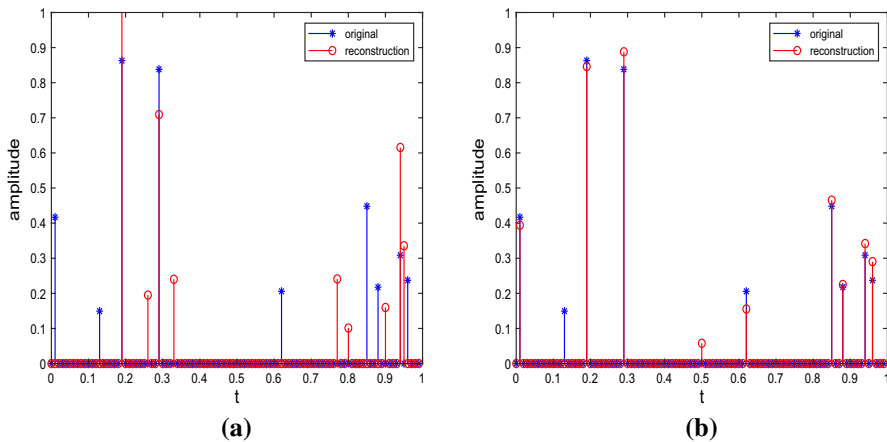


Fig. 7 FRI signal reconstruction by CS with different data length **a** $M/4$, **b** $M/3$

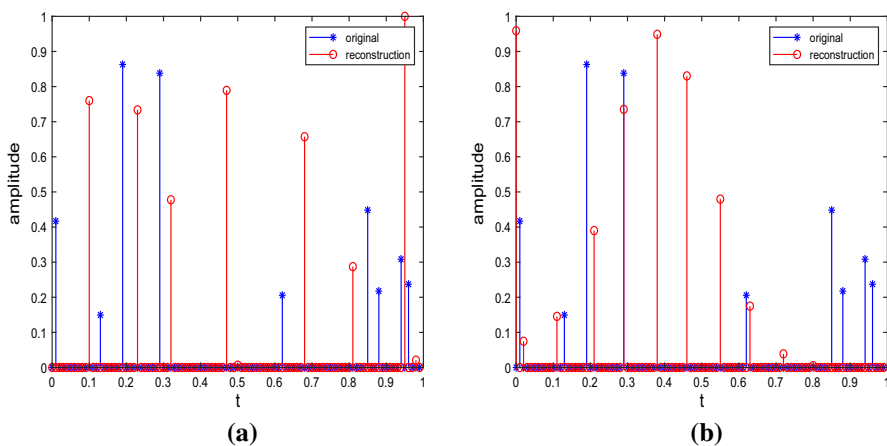


Fig. 8 FRI signal reconstruction by reproducing sampling kernels **a** E-spinles, **b** general reproducing kernel

similar performance for signal in (25), the more flexible parameters of the proposed method make it good for signal sampling and reconstruction.

Experiment 2: In order to further demonstrate the superior performance of proposed FRI-based method, this experiment aims to reconstruct other kinds of FRI-based signals. First, a time shift in LCTD is defined as

$$\mathbf{T}_{t_0}^A[x(t)] = \frac{1}{\sqrt{j2\pi b}} x(t - t_0) e^{-j\frac{a}{b}t_0(t-t_0)} \quad (26)$$

where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is parameter matrix of LCT. When $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, (26) is time shift in FTD.

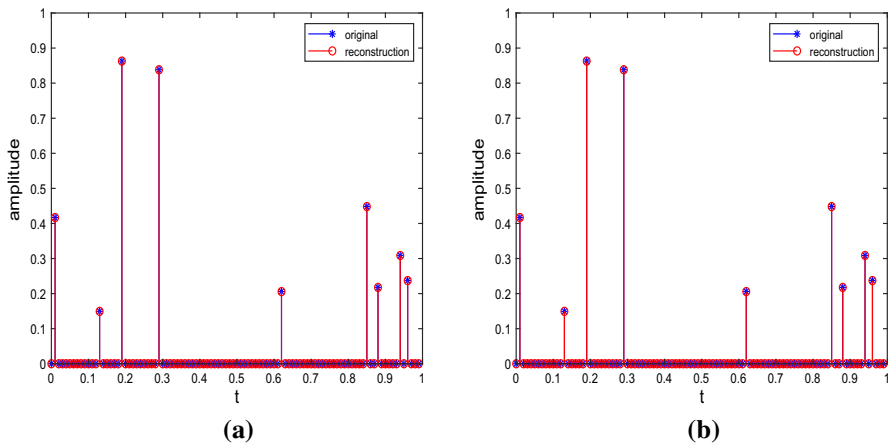


Fig. 9 FRI signal reconstruction by Sinc-of-class sampling kernels **a** traditional kernel, **b** proposed kernel

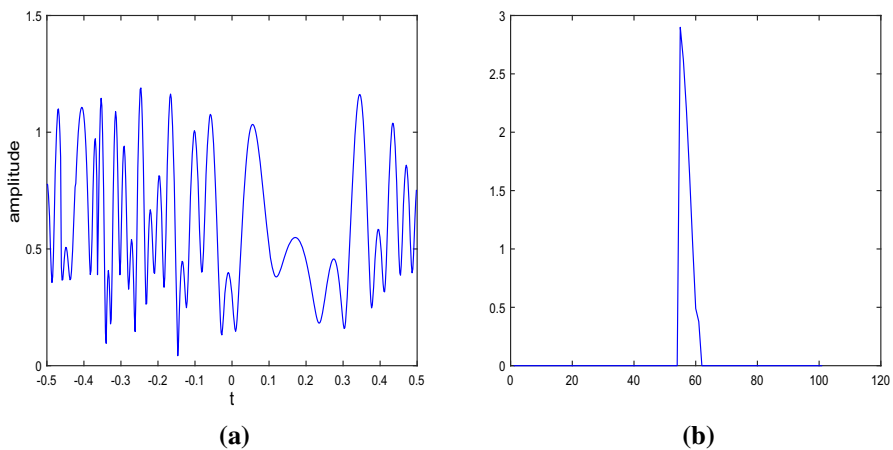


Fig. 10 FRI signal and low-pass data **a** in time domain, **b** low-pass samples in LCTD

Suppose that the signal $x(t) = e^{j\alpha t^3/3}$, $\alpha = 128$, $t \in [-0.5, 0.5]$ is via the system with $L = 3$ time shifts,

$$\sum_{l=0}^{L-1} \mathbf{T}_{t_l}^A[x(t)] = \frac{1}{\sqrt{j2\pi b}} \sum_{l=0}^{L-1} x(t - t_l) e^{-j\frac{\alpha}{b} t_l(t-t_l)} \quad (27)$$

with $\{t_l\}_{l=0}^2 = [0.0391, 0.0781, 0.1367]$, which is displayed in Fig. 10a. According to proposed FRI-based sampling method, the low-pass result in LCTD is shown in Fig. 10b.

The shift signal will be reconstructed after estimating the time shift, and the reconstruction results are shown in Fig. 11. From Fig. 11a, it is found that the traditional FRI-based method in FTD fails to reconstruct signal in (27), for which traditional

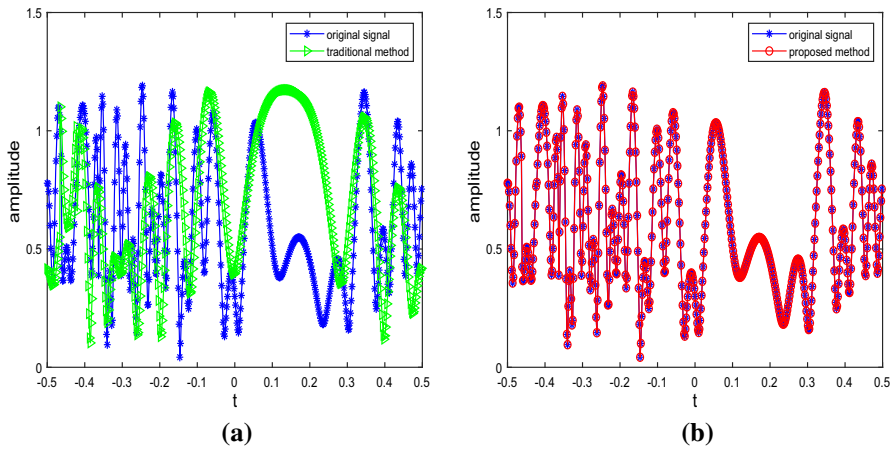


Fig. 11 FRI signal reconstruction **a** traditional method, **b** proposed method

method does not recover time shift location. The perfect reconstruction in Fig. 11b is obtained by proposed method. The LCT is the general integral transform with more free parameter, including the FRFT and the FT, which provides more refined information of signal and perform well in non-stationary signal. Thus, our proposed method has wider prospects in practical environments.

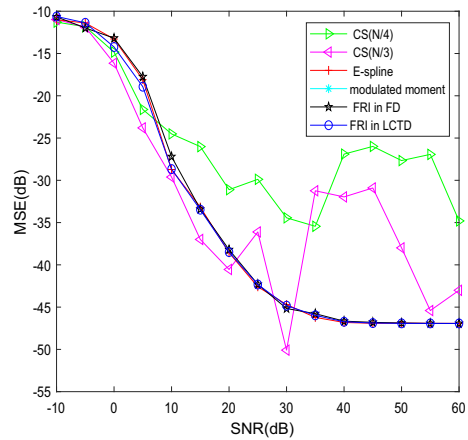
It is worth noting that the optimal parameter matrix A is usually determined by signal characteristics. Hence, it is also important to acquire and illustrate the refined characteristics of signal, which can be obtained by some optimization algorithms.

4.2 Error Analysis

It is important for proposed method to evaluate the reconstruction error in the noisy environment. Embraced into white Gaussian noise, this subsection hereby will employ MSE as a criterion to measure reconstruction error. The contrastive methods are included: CS with different observation data ($M/4$ and $M/3$), and the FRI-based methods based on reproducing sampling kernels and SoS sampling kernel in different domains (FTD and LCTD). The experiments are tested by 500 Monte Carlo simulations, and the time location results are shown in Fig. 12.

From Fig. 12, it can be seen that the errors produced by CS are unstable at some SNR values though the overall trend of MSE goes down when SNR increases, where the result based on $M/3$ data is better than $M/4$ data but with great data cost. The inherent error of CS may be caused by the grid effect and data deficiency, which causes that MSE of time location does not decrease while the SNR increase. The four sub-Nyquist FRI-based sampling methods present stable downward tendencies as the SNR increases. Among these results, the method based on the general reproducing kernel has better robust performance. The reconstruction errors of other three methods fall off quickly in the SNR range $[0, 10]$ dB, and the proposed method is more robust than the other FRI-based methods, for which the energy of signal begins to exceed

Fig. 12 MSE of time location reconstruction



that of noise. From the view of the overall SNR, the FRI-based sub-Nyquist sampling methods present similar and better reconstruction performances in noisy environments especially for that $\text{SNR} > 10$ dB. Even so, it must be mentioned that our proposed method can be applied to wider range of application scenarios due to the flexibility of parameter matrix.

4.3 Potential Application

For the proposed sub-Nyquist sampling and reconstruction method, this subsection will explore a potential application on super-resolution TDE, which has great importance in many applications including radar target location and communication channel estimation [7, 15].

Let transmit signal be the normalized linear frequency modulated (LFM) pulse, which is

$$s(t) = \text{rect}\left(\frac{t}{T_p}\right) e^{j\gamma t^2/2} \quad (28)$$

where $\text{rect}\left(\frac{t}{T_p}\right) = \begin{cases} 1, & -\frac{T_p}{2} \leq t \leq \frac{T_p}{2} \\ 0, & \text{otherwise.} \end{cases}$, $\gamma = B/T_p$ is the chirp rate of LFM signal with the bandwidth B and pulse duration T_p .

Then, the received signal from L targets can be represented as

$$r(t) = \sum_{l=0}^{L-1} a_l s(t - t_l) e^{-j\frac{a}{b} t_l(t-t_l)} \quad (29)$$

where $\text{FRI } \rho = 2L/T_p$.

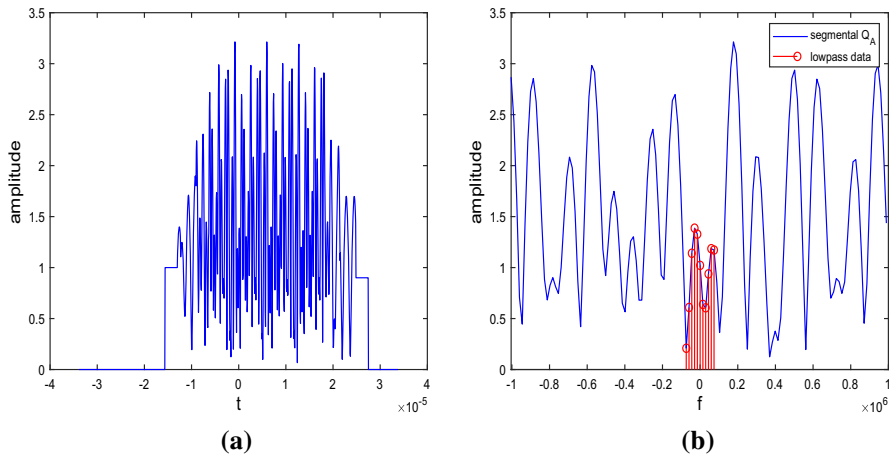


Fig. 13 Received signal and sub-Nyquist samples **a** signal, **b** low-pass samples

For (29), the optimal LCTD can be determined by $\frac{a}{b} = -\gamma$. Applying the LCT with parameter matrix A

$$\begin{aligned} R_A(u) &= \frac{1}{\sqrt{j2\pi b}} \int_{-\infty}^{\infty} r(t) e^{j(\frac{a}{2b}t^2 - \frac{2}{b}tu + \frac{d}{2b}u^2)} dt \\ &= \frac{1}{\sqrt{j2\pi b}} \sum_{l=0}^{L-1} S_A(u) a_l e^{-j\frac{a}{2b}t_l^2} e^{jut_l/b} \end{aligned} \quad (30)$$

where $S_A(u)$ is the LCT spectrum of the transmitted signal with parameter matrix A , the corresponding frequency $u = [-Bb/2, Bb/2]$. Using the proposed sampling kernel SoS_A in LCTD, the $2M_A + 1$ samples $R_A[m] = R_A[mu_0]$, $S_A[m] = S_A[mu_0]$, $m \in [-M_A, M_A]$ are reproduced. Then,

$$Q_A[m] = \frac{R_A[m]}{S_A[m]} = \sum_{l=0}^{L-1} q_l u_l^m \quad (31)$$

where $Q_A[m]$ shares the form of (1) with $q_l = \frac{1}{\sqrt{j2\pi b}} a_l e^{-j\frac{a}{2b}t_l^2}$, $u_l = e^{ju_0 t_l/b}$. Subsequently, the above TDE problem can be solved by proposed Algorithm 1.

To clarify the potential application of the proposed FRI-based method on super-resolution TDE, an echo example with $L = 5$ targets is considered. The transmitted signal is $s(t) = \text{rect}(\frac{t}{T_p}) e^{j\pi\gamma t^2}$ with bandwidth $B=7.56$ MHz and pulse width $T_p=16.9 \mu\text{s}$. Figure 13a shows the received signal, in which time delays are $\{t_l\}_{l=0}^4 = [20 \ 60 \ 70 \ 120 \ 160]T_s$ with discrete interval T_s and amplitudes are $\{a_l\}_{l=0}^4 = [1 \ 0.4 \ 0.5 \ 0.8 \ 0.9]$. According to Theorem 1, $N = 2M_A + 1 = 11$ samples of $R_A(u)/S_A(u)$ are obtained. $Q_A[m]$, $m = 0, \dots, N - 1$ are shown in red circle

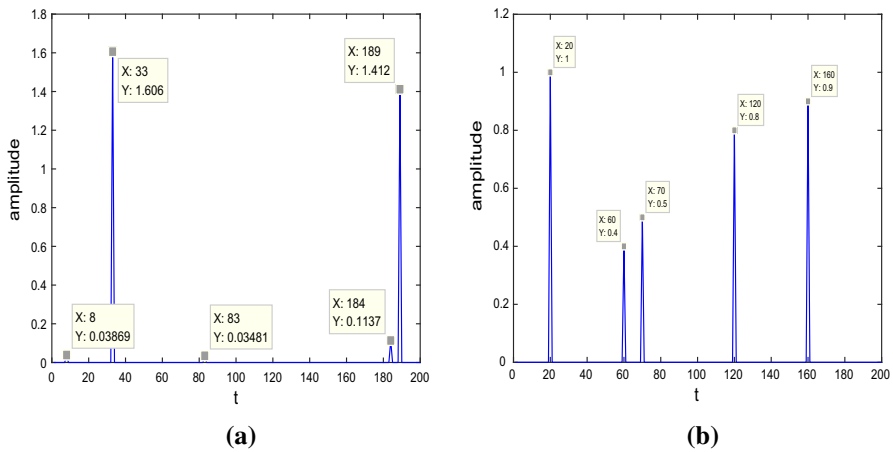


Fig. 14 Super-resolution TDE by FRI-based sampling methods **a** traditional method, **b** proposed method

in Fig. 13b. For the sake of clarity, the blue line in Fig. 13b is the part of the whole continuous $R_A(u)/S_A(u)$.

The TDE results in Fig. 14 are obtained via the traditional FRI-based sampling method and proposed method, respectively, among which Fig. 14b shows better accuracy of time delays. Based on the proposed FRI-based sub-Nyquist sampling method in LCTD, we will conduct more in-depth research and applications in future work.

5 Conclusion

In this paper, we propose a novel sub-Nyquist sampling and reconstruction method in LCTD. A novel compact support sampling kernel, SoS_A, is first designed in LCTD to acquire sub-Nyquist samples. Then, the corresponding sampling theorem is derived. Associated with the AF and LS methods, the reconstruction algorithm is summarized. Numerical experiment demonstrates the correctness and effectiveness of signal reconstruction. In addition, comparison experiment results validate that our proposed method has superior reconstruction performance than that of other compared methods. Finally, the proposed FRI-based sampling method in LCTD can be applied to TDE with super resolution, which verifies that the proposed method has potential to improve resolution of received signal. We will study more sampling theory and wider practical environments of the proposed method in our future work.

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Availability of data and materials Not applicable.

Declarations

Conflict of interest The authors declare no conflicts of interest.

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